

Practice 3. LABVIEW

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Practice 3a. GPIB INSTRUMENTS CONTROL: ARBITRARY FUNCTION GENERATOR (HP33120A)

PART 1. Devices identification connected to bus by MAX

Instrument	Manufacturer	Model	Primary address
Function generator	Hewlett-Packard	33120A	17
Oscilloscope	Rigol Technologies	DS2102A	172.16.148.63

GENERATOR

33120A "GPIB0::17::INSTR" - Measurement & Automation Explorer

File Edit View Tools Help

My System

Data Neighborhood

Devices and Interfaces

ASRL1::INSTR "COM1"

ASRL10::INSTR "LPT1"

NI USB-4431 "Dev3"

NI PCI-6221 "Dev2"

NI PCI-GPIB "GPIB0"

33120A "GPIB0::17::INSTR"

Network Devices

DS2102A "TCPIP0::b221-osd-13.gnd.upv.es::inst0::INSTR"

SDG2122X "TCPIP0::b221-gen-13.gnd.upv.es::inst0::INSTR"

TCPIP0::b221-fad-13.gnd.upv.es::inst0::INSTR

Scales

Software

Remote Systems

Save Refresh

Open VISA Test Panel

Communicate with Instrument

Interactive Control

Settings

Name

Vendor HEWLETT-PACKARD

Model 33120A

Serial Number 0

Firmware Version 8.0-5.0-1.0

Status Present

Primary Address 17

VISA Resource Name GPIB0::17::INSTR

33120A "GPIB0::17::INSTR" - Measurement & Automation Explorer

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Vendor HEWLETT-PACKARD

Model 33120A

Serial Number 0

GPIB0::17::INSTR - VISA Test Panel

Configuration Input/Output Advanced

Basic I/O Line Control

Select or Enter Command *IDN?\n

*IDN?\n

Bytes to Read 1024

Write Query Read Read Status Byte Clear

View mixed ASCII/hexadecimal

3: Write Operation (*IDN?\n)

Return Count: 6 bytes

4: Read Operation

Return Count: 37 bytes

HEWLETT-PACKARD,33120A,0,8.0-5.0-1.0\n

Copy to Clipboard Clear Buffer

Return Data

Read Operation

No Error

GPIB0::17::INSTR - VISA Test Panel

Configuration Input/Output Advanced

Basic I/O Line Control

Select or Enter Command OUTPUT:LOAD?

OUTPUT:LOAD? Bytes to Read 1024

Write Query Read Read Status Byte Clear

View mixed ASCII/hexadecimal

22: Write Operation (OUTPUT:LOAD?)
Return Count: 12 bytes

23: Read Operation
Return Count: 13 bytes
+5.00000E+01\n

Copy to Clipboard Clear Buffer

Return Data
Read Operation
No Error

GPIB0::17::INSTR - VISA Test Panel

Configuration Input/Output Advanced

Basic I/O Line Control

Select or Enter Command OUTPUT:LOAD\sINF

OUTPUT:LOAD\sINF Bytes to Read 1024

Write Query Read Read Status Byte Clear

View mixed ASCII/hexadecimal

23: Read Operation
Return Count: 13 bytes
+5.00000E+01\n

24: Write Operation (OUTPUT:LOAD\sINF)
Return Count: 15 bytes

Copy to Clipboard Clear Buffer

Return Data
Write Operation (OUTPUT:LOAD\sINF)
No Error

GPIB0::17::INSTR - VISA Test Panel

Configuration Input/Output Advanced

Basic I/O Line Control

Select or Enter Command OUTPUT:LOAD?

OUTPUT:LOAD? Bytes to Read 1024

Write Query Read Read Status Byte Clear

View mixed ASCII/hexadecimal

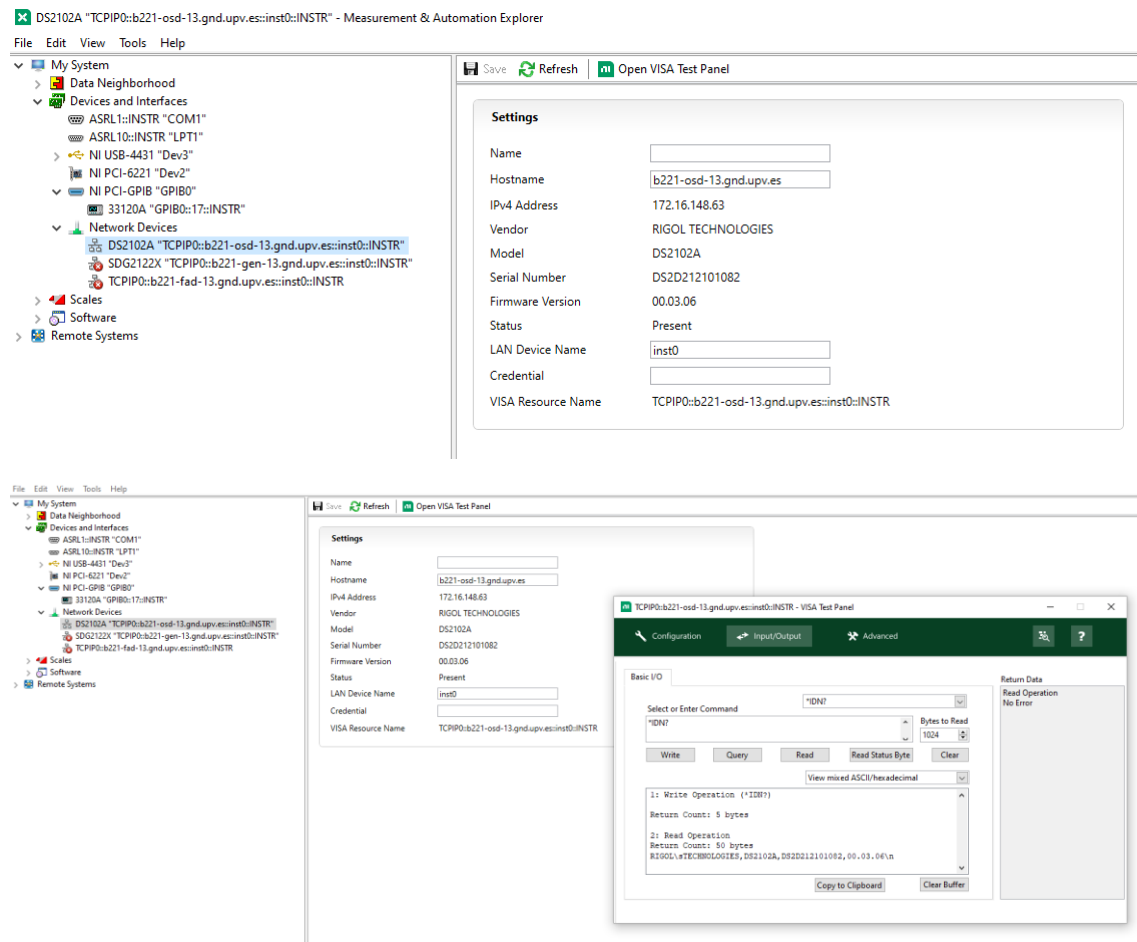
25: Write Operation (OUTPUT:LOAD?)
Return Count: 12 bytes

26: Read Operation
Return Count: 13 bytes
+9.90000E+37\n

Copy to Clipboard Clear Buffer

Return Data
Read Operation
No Error

OSCILLOSCOPE



PART 2. Control of Arbitrary Function Generator from the program VISA Interactive Control.

1,2. Commands used:

*IDN?

*RST

*IDN?

OUTPUT:LOAD?

OUTPUT:LOAD INF

APPL:SIN 5.0e+3,3.0,-2.5

APPL:DC DEF, DEF,-2.5

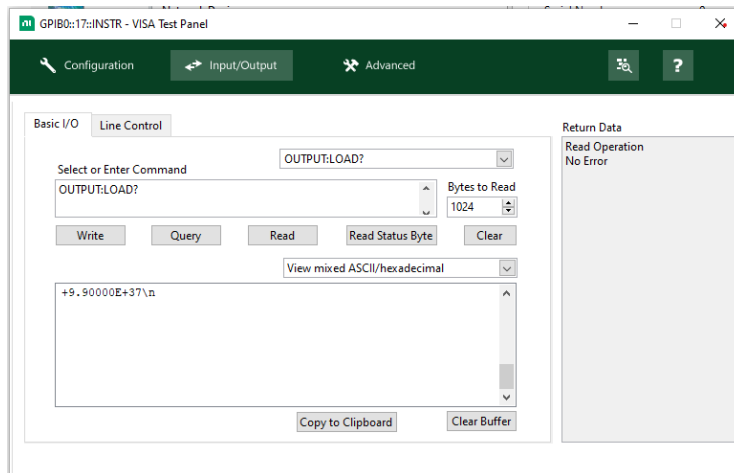
APPL:NOIS DEF,5.0,2.0

3. Setting the output impedance:

OUTPUT:LOAD? +5.00000E+01\n

OUTPUT:LOAD INF

OUTPUT:LOAD? +9.90000E+37\n



4. COMANDS for different signals:

APPL:SIN 1.0e+3,1.0,0

APPL:SQU 0.5e+3,3.0,1.5

PULS:DCYC 70

APPL:TRI 20e+3,0.5,0

APPL?

FUNC:USER CARDIAC

FUNC:SHAP USER

FUNC:USER SINC

FUNC:SHAP USER

5. COMANDS for signal modulation

APPL:SIN 1.0e+3,1

AM:INT:FUNC SIN

AM:INT:FREQ 10.0e+3

VOLT 1.0

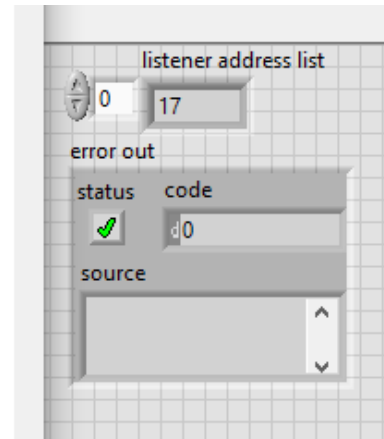
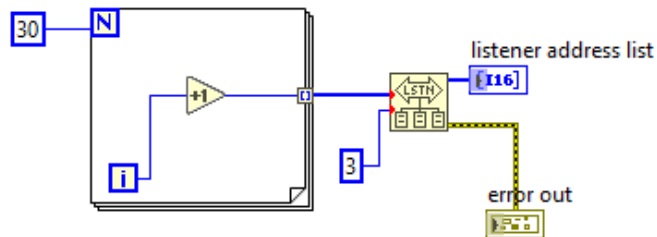
AM:DEPT 30

AM:STAT ON

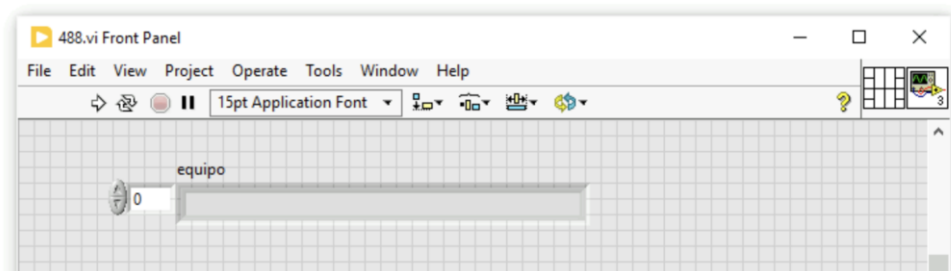
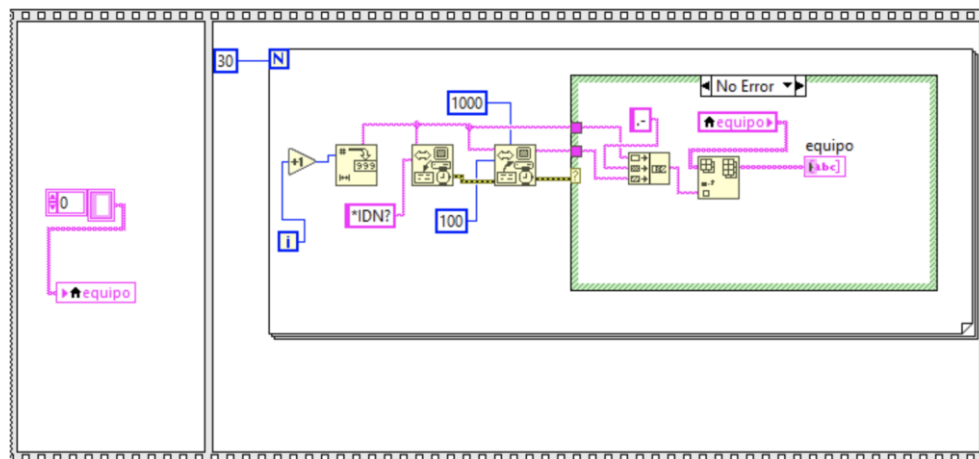
Devices identification connected to bus by LabVIEW

LabVIEW GPIB

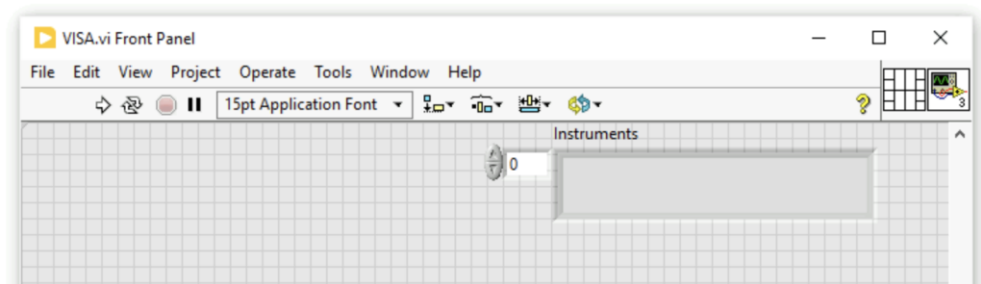
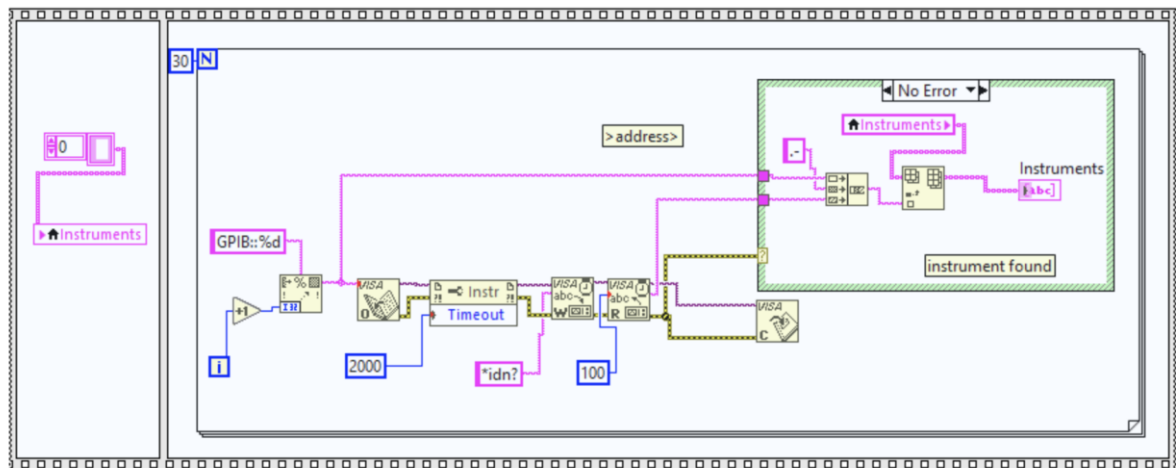
FindLstn listener



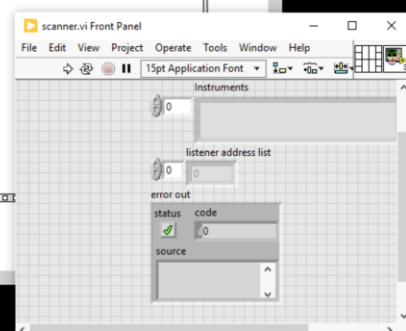
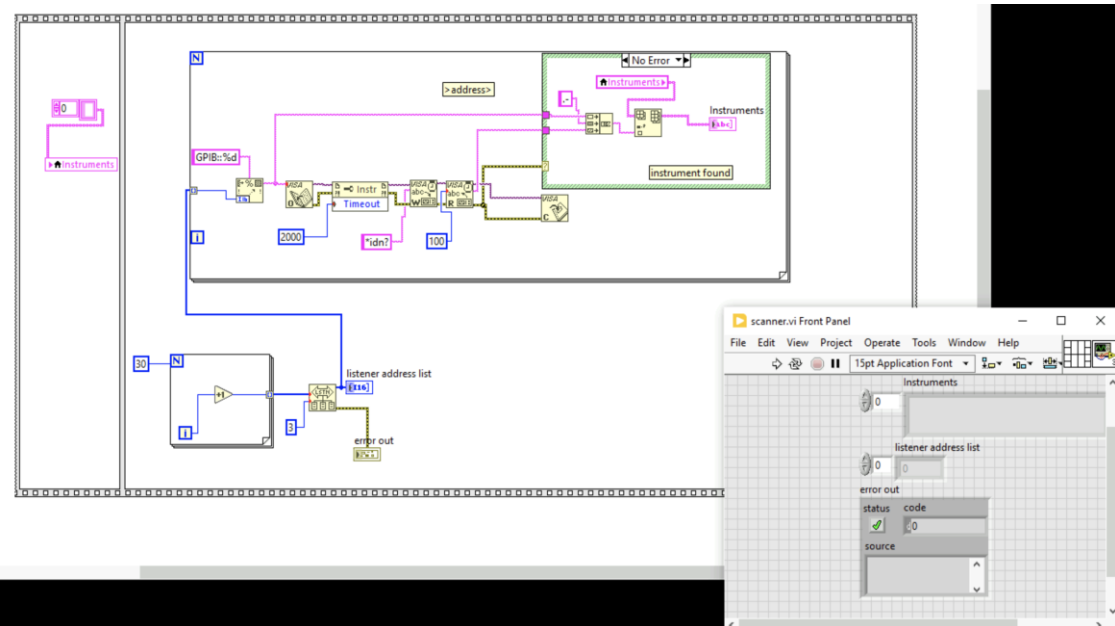
- **488 functions**



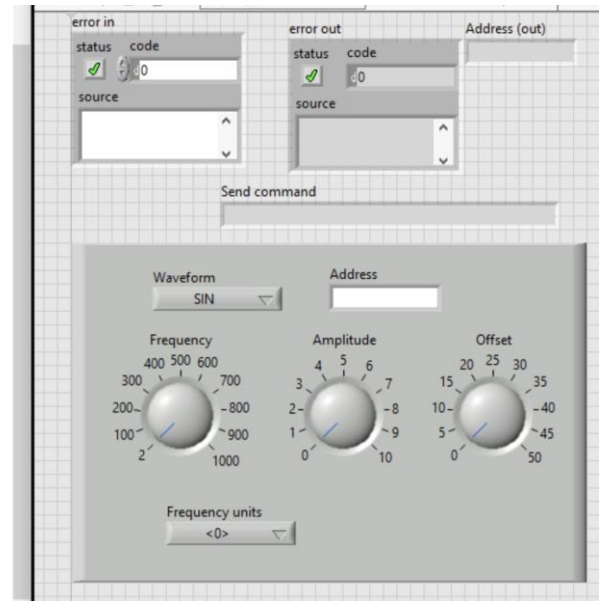
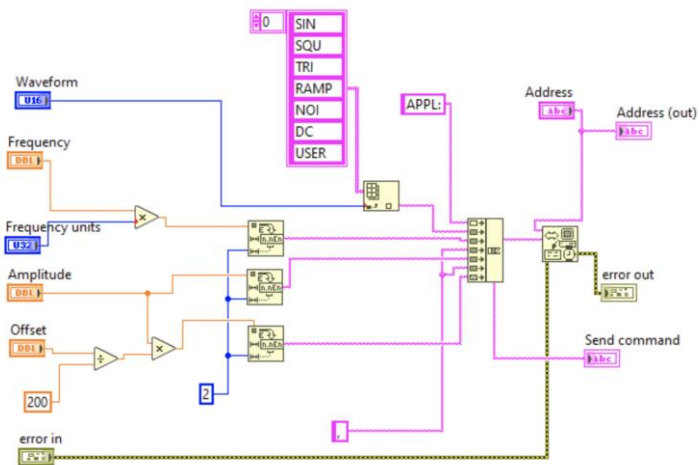
- **VISA functions**



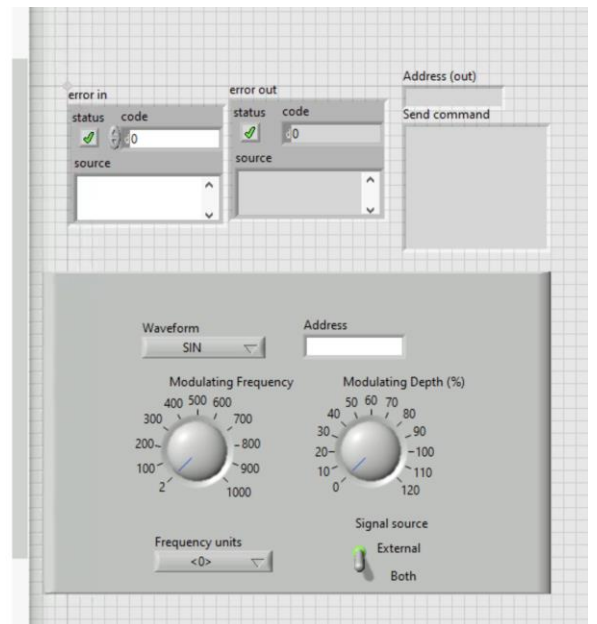
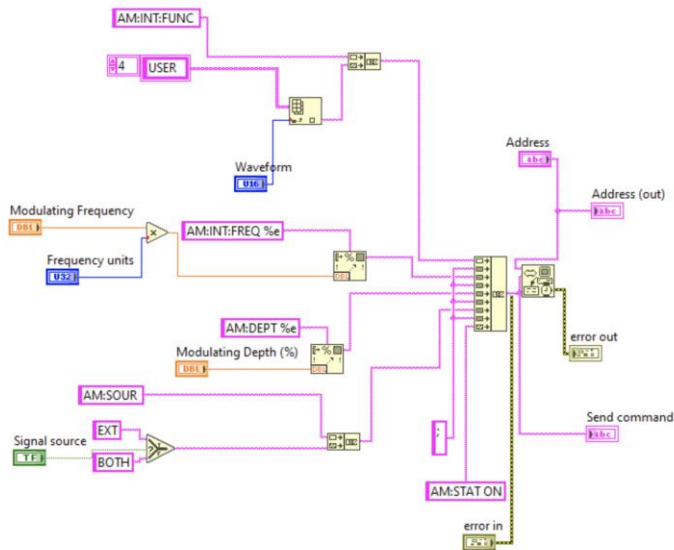
- Optimized scanner: scanner.vi



A) Generation basic waves



B) AM Signal Generation



Complete Generator

